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Data Handling & Visualization

Q. Explain descriptive vs inferential statistics with examples.

Descriptive statistics

- Descriptive statistics deal with summarizing and visually presenting the main features of a dataset.
- They help in understanding what the data shows without making conclusions beyond data.
- Common descriptive measures include mean, median, mode, range, variance & standard deviation.
- Descriptive statistics are represented using histograms bar charts, pie charts, box plots, and line graphs.
- For example, a histogram showing the distribution of students, exam scores or a box plot visualizing salary spread are descriptive.

Inferential statistics

- Inferential statistics focus on drawing conclusions or making predictions about a population based on a sample.
- They involve hypothesis testing, confidence intervals regression, and correlation.
- In visualization, inferential statistics are supposed to using scatter plots with regression lines, error bars & confidence interval plots.
- For example, a scatter plot with a fitted line to infer the relationship between study hours and exam scores represents inferential statistics.

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Q. Explain random variables & their role in statistics.

- A random variable is a variable whose values depend on the outcome of a random experiment.
- It quantatively represents uncertainity in data.
- Random variables are classified into discrete & continuous types.
- Discrete random variables take countable values, such as the number of customers visiting a store, while continuous random variables take values over a range, such as height or temperature.
- In data visualization, random variables are visualized to understand their distribution & behaviour.
- Discrete random variables are commonly visualized using bar charts, while continuous random variables are visualized using histograms, density plots & box plots.
- Random variables from the foundation of probability distributions, which are essential for interpreting visual patterns & variability in data.

Q. Discuss bivariate & multivariate visualization techniques.

- Bivariate visualization is used to analyze the relationship between two variables.
- Common techniques include scatter plots, line charts, grouped bar charts, and box plots.
- For example, a scatter plot showing height versus weight helps identify correlation patterns between two variables.

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- Multivariate visualization involves more than 2 variables and helps identify complex patterns & interactions.
- Techniques include scatter plot matrices, parallel cordinate plots, heatmaps, bubble charts & 3D plots.
- For example, a heatmap showing correlation among multiple features or a bubble chart representing sales, profit & region simultaneously are multivariate visualizations.
- These techniques are essential in exploratory data analysis to uncover hidden trends.

Q. Explain histogram matrix and scatter plot matrix in detail.

- A histogram matrix is a grid of histograms where each histogram represents the distribution of 1 variable in a dataset.
- It is mainly used to understand the spread, skewness and shape of multiple variables simultaneously.
- Histogram matrices help compare distributions across features & detect outliers visually.
- A scatterplot matrix is a grid of scatter plots where each cell shows the relationship between a pair of variables.
- The diagonal usually contains histograms or density plots of individual variables.
- Scatter plot matrices are widely used to detect correlations, clusters, and linear or non-linear relationships among multiple variables.
- They are particularly useful in multivariate data analysis & feature selection.

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Q. Explain variables, functions, and built-in functions in R with examples.

- In R, variables are used to store data values such as numbers, vectors, or datasets.
- For example, x <- c(10, 20, 30) assign values to a variable used for plotting.
- Functions in R are reusable blocks of code that perform specific tasks.
- User-defined functions allow customization, such as creating a function to normalize data before visualization.
- Built-in functions in R simplify data visualization and analysis.
- Common examples include mean(), sd(), summary() and plotting functions like plot(), hist(), boxplot() and pairs().
- hist(data) creates a histogram while, plot(x, y) generates scatter plot.

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